Gender Identity and Ideology Shape Perceptions of Masculinity in Male Speech

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Perceiving gender

Gender and /s/:

- /s/ acoustics → gender *identity* and *performance*
  - High CoG /s/ → female voices, femininity, non-normative masculinity
    - (e.g., Strand & Johnson 2006, Munson 2007)

Matched guise and perceiving social meaning:

- Suggest that listeners hold consistent perceptions of fronted (high CoG) /s/ across speakers (Boyd et al. 2021, Campbell-Kibler 2011)



Tendency to use speakers whose voices are “unmarked” w.r.t. gender



Assuming homogeneous listening population behavior

How stable is the indexical connection between /s/ and perceived masculinity?

How do listeners' gender identities **and** gendered beliefs influence their judgments about the masculinity of multiple male voices?

Experimental Design

Matched-guise “inspired” experiment

Within-subjects design (2x4 design):

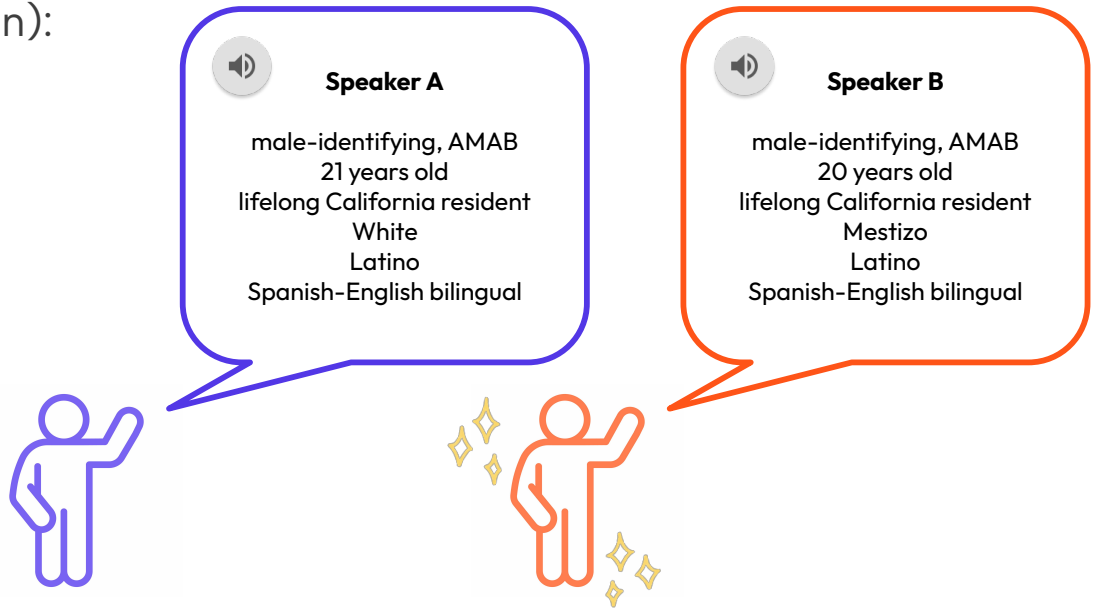
- Speaker: 2 speakers

Experimental Design

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- Speaker: 2 speakers
- /s/ conditions:
  - **Front /s/**
  - **Mid /s/**
  - **Back /s/**
  - **Control (no /s/)**

Participants heard multiple combinations of each speaker : /s/ condition

Who's listening?

102 cisgender females

74 gender-diverse

93 cisgender males

Female, woman, cis female, reluctant woman

Agender, dual-gender, gender-fluid, non-binary/NB, two-spirit, queer, trans*, NB/trans masc, NB/genderqueer

Male, man, M

Experimental Design

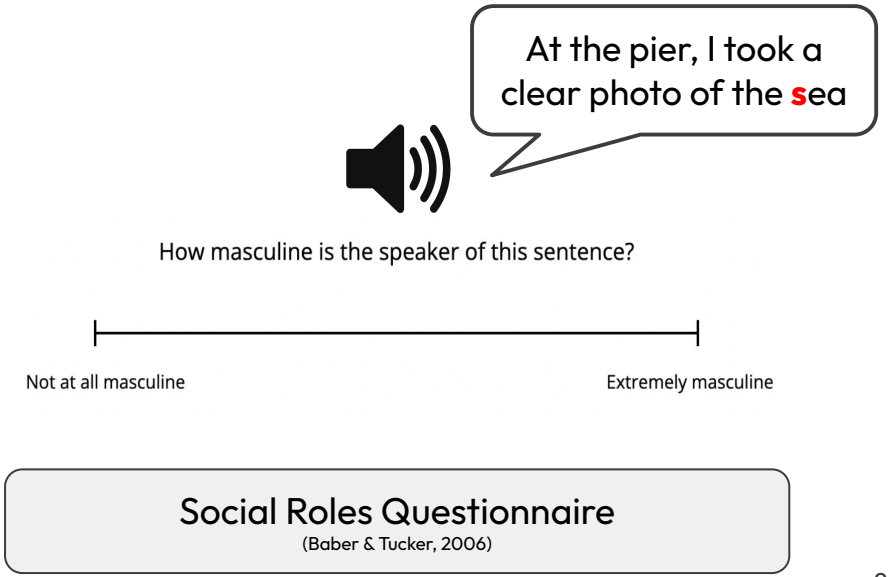
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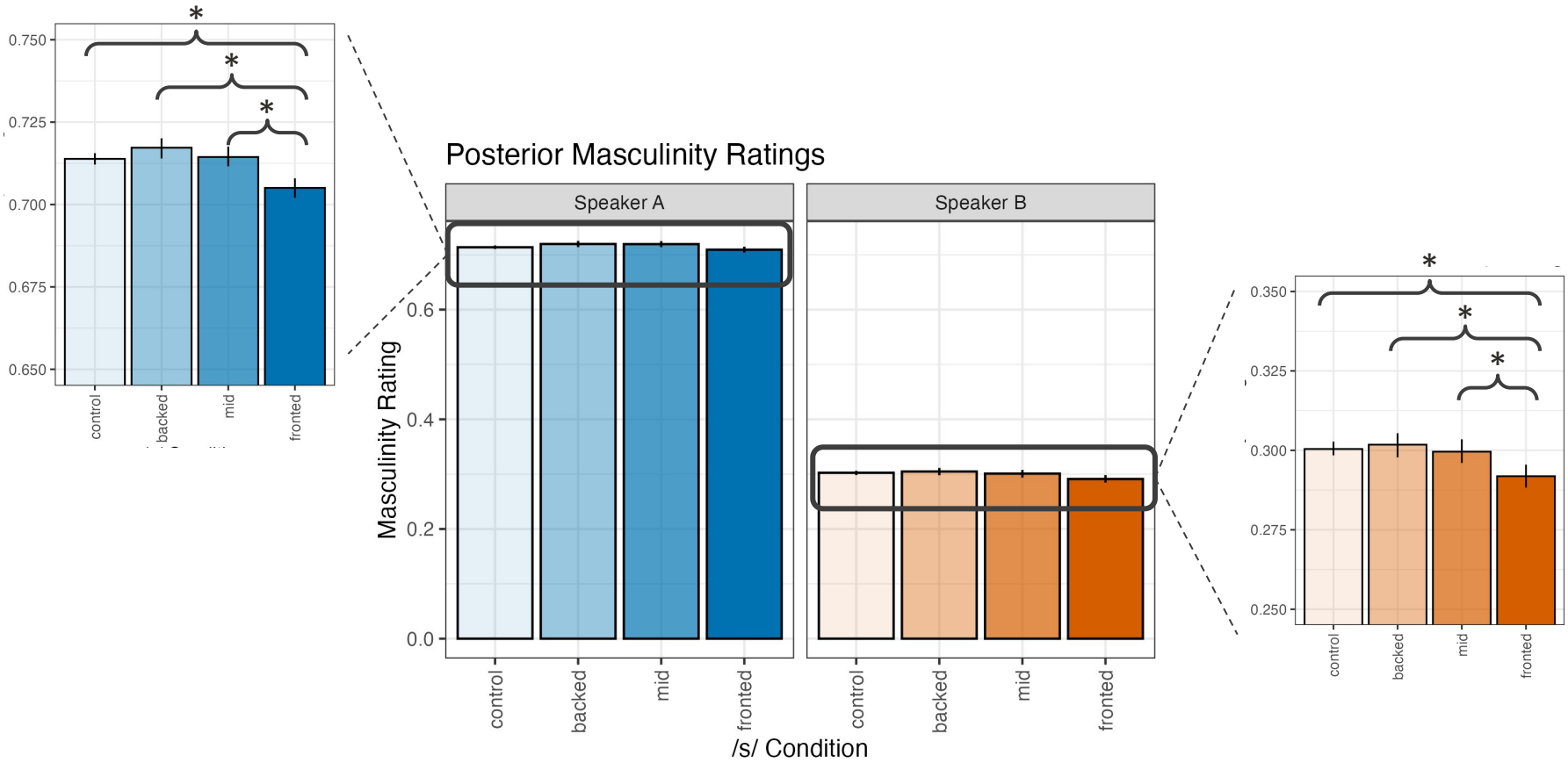
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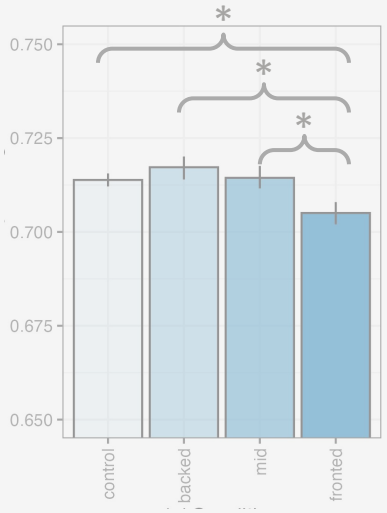
Participants: 269 monolingual English speakers in the US (via Prolific)

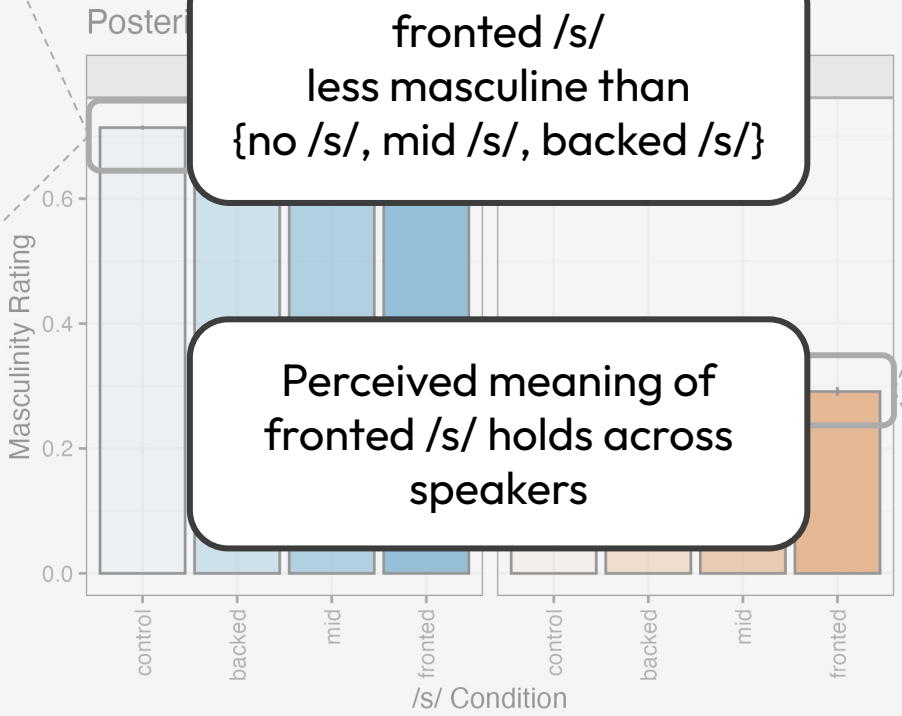


Social Roles Questionnaire
(Baber & Tucker, 2006)

Results

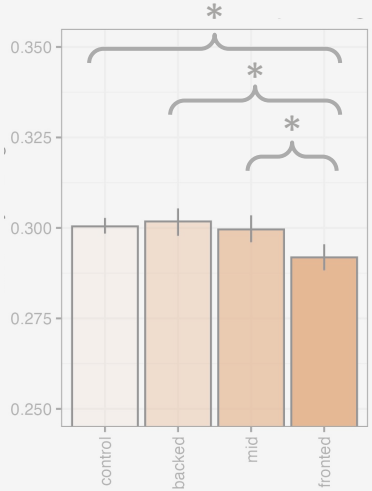

Posterior Masculinity Ratings
Speaker A
Speaker B
Masculinity Rating
0.0
0.2
0.4
0.6
control
backed
mid
fronted
/s/ Condition
0.650
0.675
0.700
0.725
0.750
0.250
0.275
0.300
0.325
0.350
*

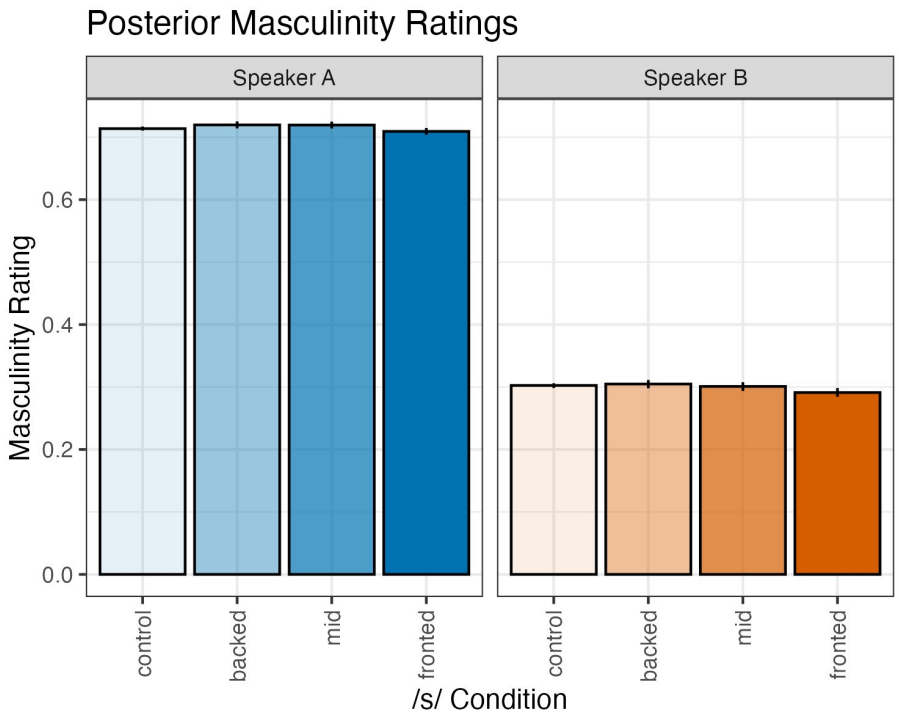




fronted /s/
less masculine than
{no /s/, mid /s/, backed /s/}

Perceived meaning of fronted /s/ holds across speakers

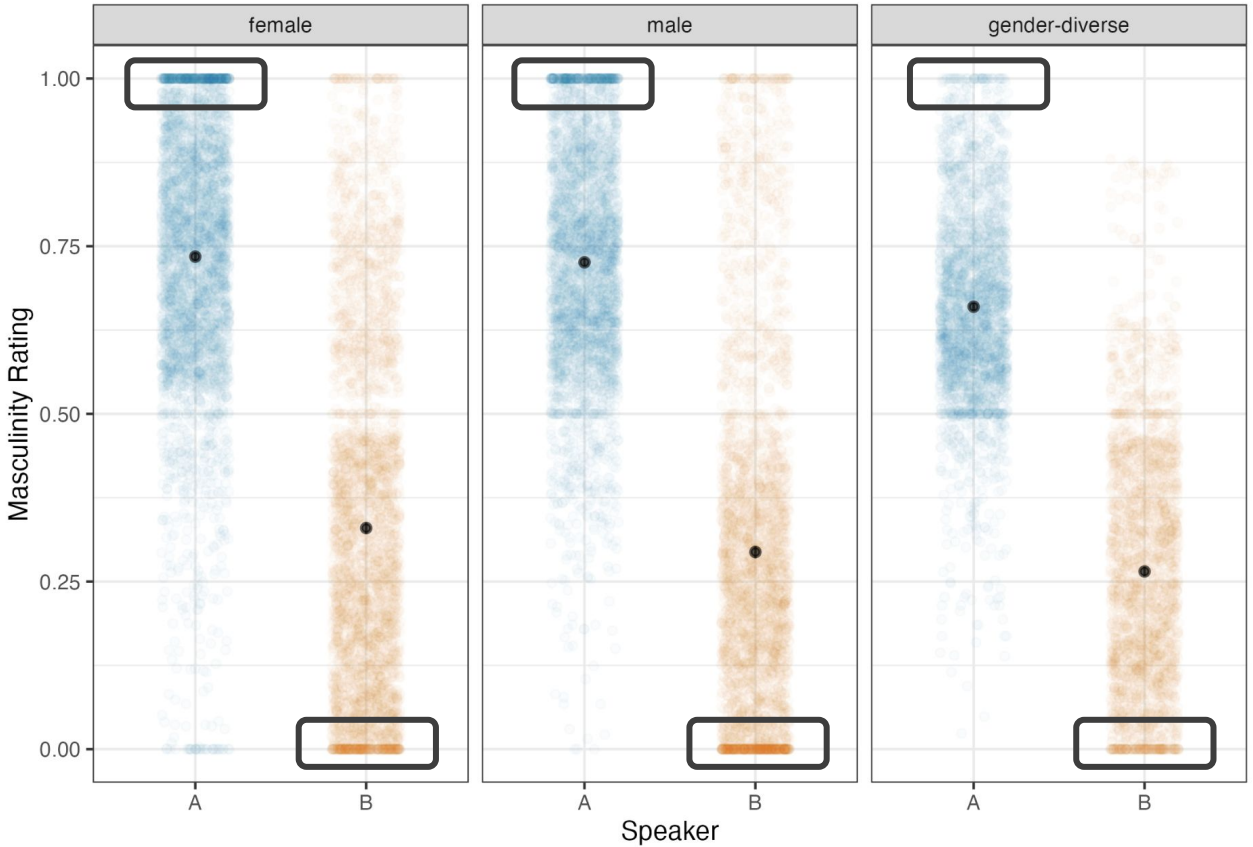




Within-utterance measurements of:

- Creak
- Mean F0
- F0 Range
- Speech Rate

Use of endpoints



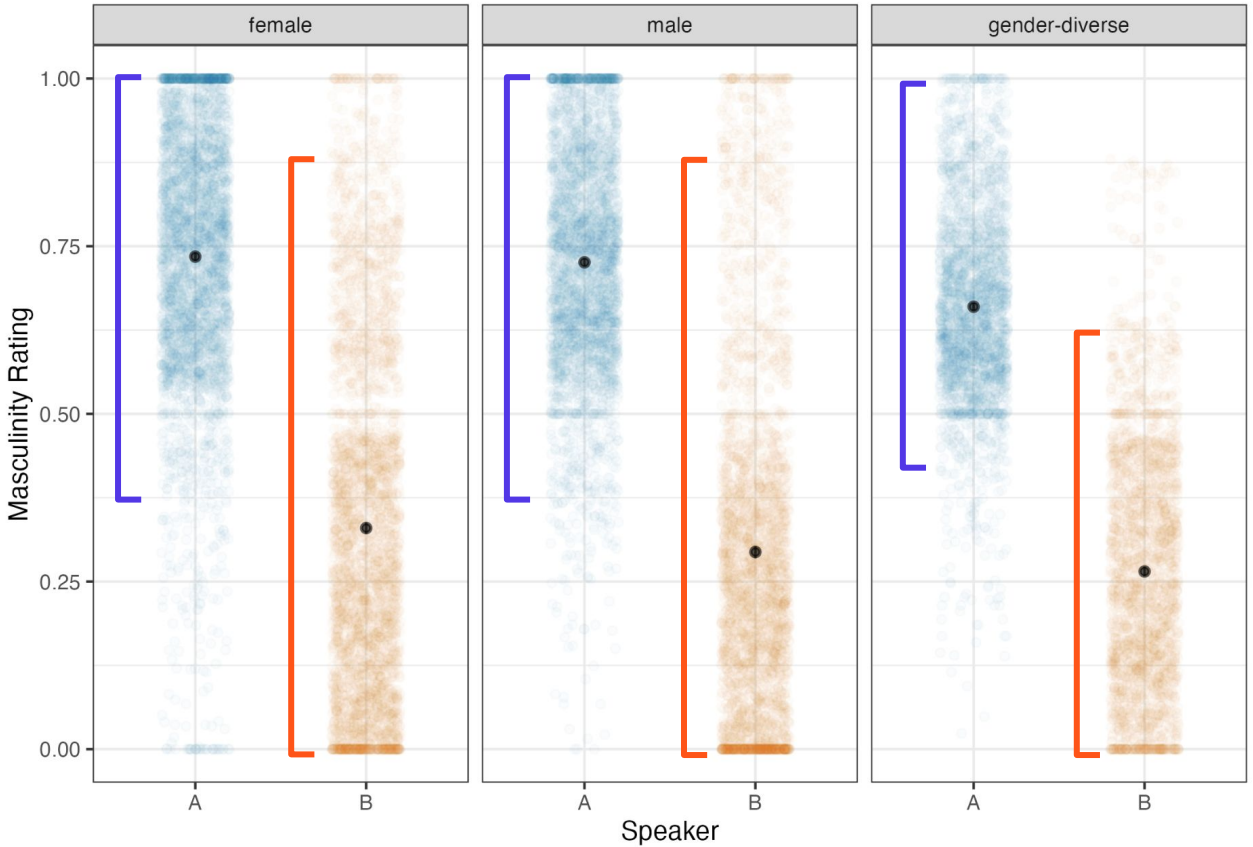
Via Zero-One Inflated Beta Regression:

Cisgender listeners use scale endpoints more than gender-diverse listeners

(post. prob > 0.99)

Gender-diverse listeners resist "absolute" positioning of both speakers

Variability of responses



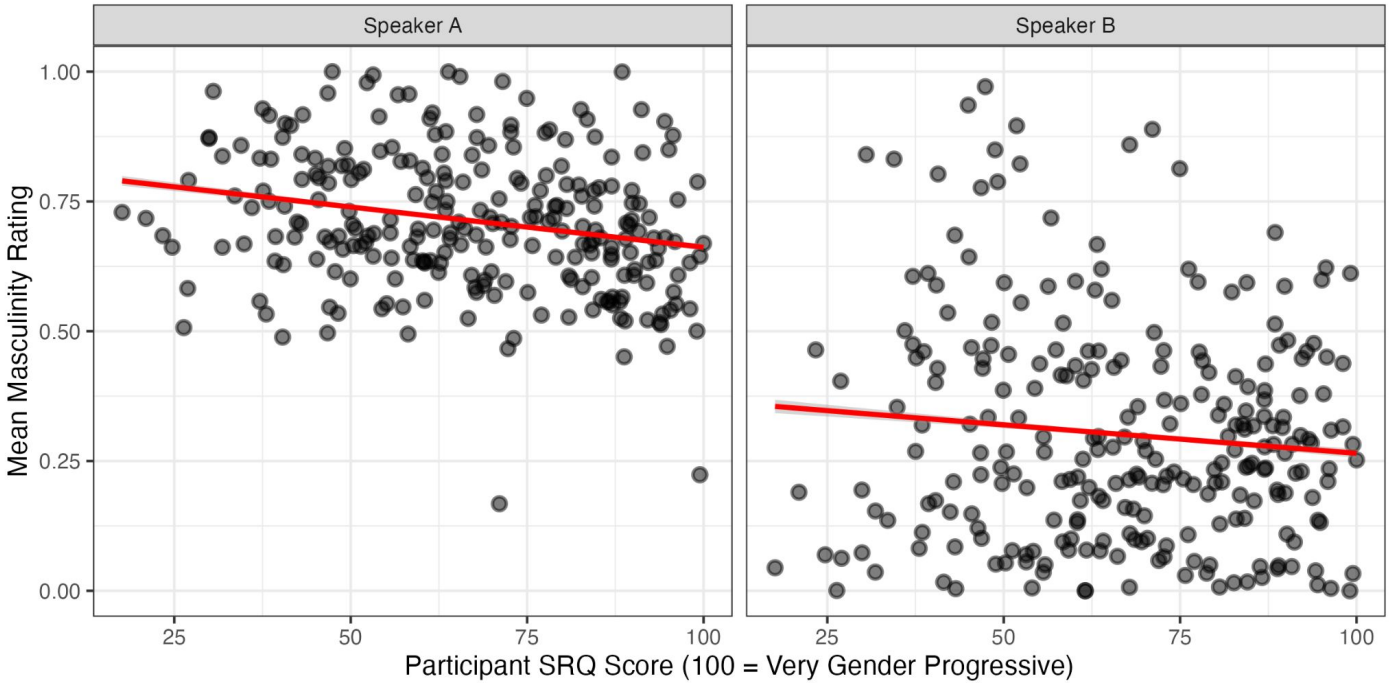
Via Zero-One Inflated Beta Regression:

Cisgender listeners vary more in their rating behavior than gender-diverse listeners

(post. prob > 0.99)

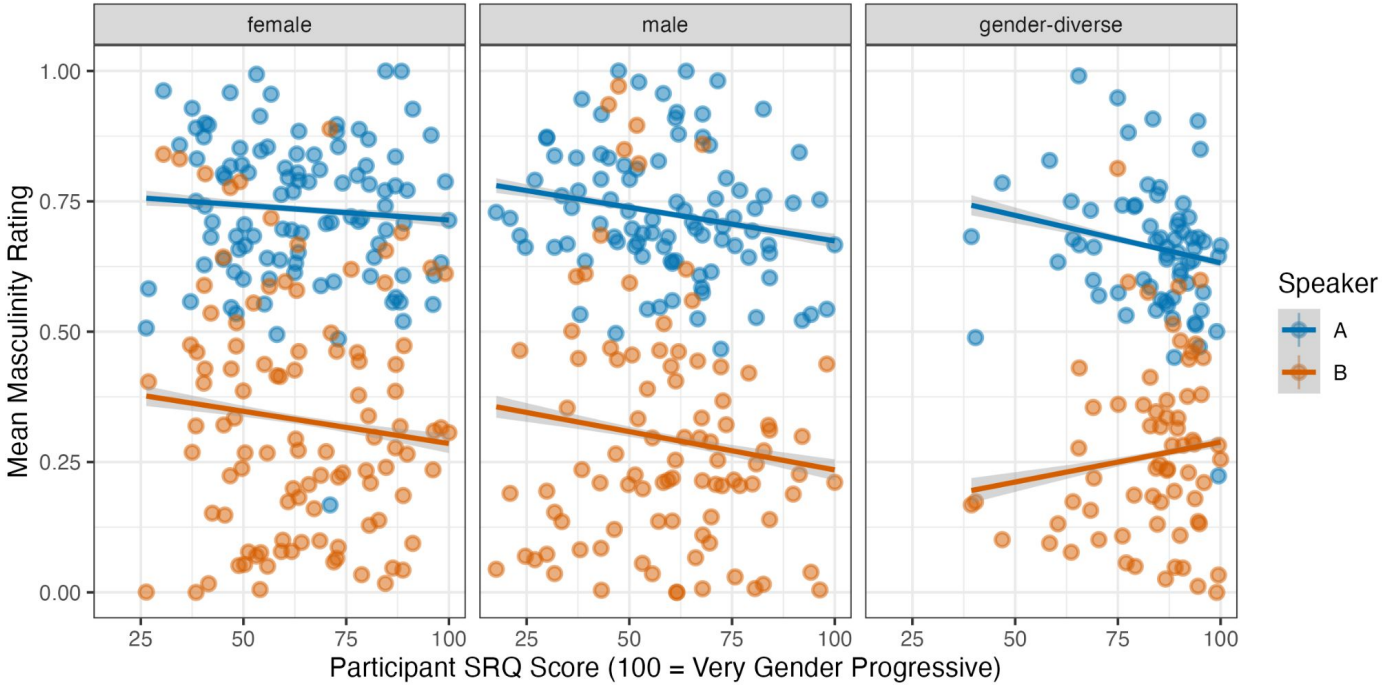
Gender-diverse listeners engage differently with the non-normative speaker

Mean masculinity ratings by listener gender ideology



"Non-masculinity" is not viewed as a negative social label by progressive listeners

What do we gain by looking at listener gender?



We cannot disentangle gender-diverse identities from progressive gender ideologies

Identity 🔗 Ideology

- Social meanings of /s/ persist across speaker and listener contexts

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- Gender diverse listeners resist broad perceptual patterns of cisgender listeners
    - GD listeners tend to use endpoints less
    - GD listeners vary less as a group in their speaker masculinity ratings
    - More progressive GD listeners rate Speaker A as less masc and Speaker B as more masc

Identity 🔗 Ideology

- Social meanings of /s/ persist across speaker and listener contexts
- Gender diverse listeners resist broad perceptual patterns of cisgender listeners
    - GD listeners tend to use endpoints less
    - GD listeners vary less as a group in their speaker masculinity ratings
    - More progressive GD listeners rate Speaker A as less masc and Speaker B as more masc
- How people engage with slow social rating tasks is highly ideological

In order to understand the perceptual impact of ideology, we must consider *whose* ideology it is

Selected References

- Agha, A. (2003). The social life of cultural value. *Words and Beyond: Linguistic and Semiotic Studies of Sociocultural Order, 23*(3):231–273.
- Boyd, Z., Fruehwald, J., and Hall-Lew, L. (2021). Crosslinguistic perceptions of /s/ among English, French, and German listeners. *Language Variation and Change, 33*(2):165–191.
- Calder, J. (2019), From Sissy to Sickening: The Indexical Landscape of /s/ in SoMa, San Francisco. *J Linguist Anthropol, 29*: 332-358.
- Campbell-Kibler, K. (2011). Intersecting Variables and Perceived Sexual Orientation in Men. *American Speech, 86*(1):52–68.
- Eckert, P. (2008). Variation and the Indexical Field. *Journal of Sociolinguistics, 12*:453–476.
- Levon, E. (2014). Categories, stereotypes, and the linguistic perception of sexuality. *Language in Society*, *43*(5), 539–566.
- Munson, B. (2007). The Acoustic Correlates of Perceived Masculinity, Perceived Femininity, and Perceived Sexual Orientation. *Language and Speech, 50(*1):125–142. Publisher: SAGE Publications Ltd.
- Munson, B., Jefferson, S., and McDonald, E. (2006). The influence of perceived sexual orientation on fricative perception. *The Journal of the Acoustical Society of America, 119*:2427–37.
- Podesva, R. J. and Van Hofwegen, J. (2016). /s/exuality in smalltown California: Gender normativity and the acoustic realization of /s/. In Levon, E. and Mendes, R. B., editors, *Language, Sexuality, and Power,* pages 168–188. Oxford University Press.
- Woolard, K. A. (2020). Language Ideology. In *The International Encyclopedia of Linguistic Anthropology* (pp. 1–21).
- Zimman, L. (2017). Variability in /s/ among transgender speakers: Evidence for a socially grounded account of gender and sibilants. *Linguistics, 55.*